

MACKENZIE, BC, Canada

WMO: 719440

Lat: 55.299N

Lon: 123.133W

Elev: 2264

StdP: 13.53

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-14.1	-30.3	1.1	-19.3	-23.1	1.7	-12.1	18.9	30.3	16.7	27.4	2.9	0	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	81.8	60.7	78.1	59.1	74.5	57.4	62.5	77.4	60.8	74.3	59.1	71.3	6.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.1	75.6	65.8	55.6	71.4	63.8	54.0	67.4	62.6	29.2	77.8	27.9	74.4	26.7	71.4	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	14.0	12.2	DB	-29.1	87.9	10.1	3.3	-36.4	90.3	-42.4	92.2	-48.1	94.0	-55.4	96.4	(4)
(5)			WB	-28.7	65.5	9.5	1.6	-35.6	66.6	-41.2	67.6	-46.5	68.4	-53.5	69.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	15.0	18.2	26.0	37.7	48.8	56.3	60.4	58.4	49.7	38.1	25.4	17.1		
	DBStd	18.83	16.30	12.62	11.47	6.54	6.89	5.45	5.11	5.68	6.24	6.54	9.96	12.94		
	HDD50	5418	1086	891	743	370	107	8	0	3	80	371	738	1021		
	HDD65	10002	1551	1311	1208	819	504	266	161	216	459	833	1188	1486		
	CDD50	930	0	0	0	1	69	198	323	265	71	3	0	0		
	CDD65	37	0	0	0	0	1	5	19	12	0	0	0	0		
	CDH74	873	0	0	0	0	38	147	392	275	21	0	0	0		
	CDH80	194	0	0	0	0	4	22	107	59	2	0	0	0		
	WSAvg	4.9	5.0	4.5	5.1	5.2	5.2	5.0	4.8	4.5	4.7	4.9	4.9	4.7		
	PrecAvg	27.0	3.1	2.2	1.9	1.0	1.8	2.5	2.6	2.1	2.0	2.6	2.7	2.9		
(15) Precipitation	PrecMax	33.0	5.4	5.4	3.7	3.4	5.0	5.0	5.2	3.7	3.9	4.3	4.9	5.2		
	PrecMin	19.6	1.0	0.9	0.4	0.1	0.5	0.5	0.6	0.4	0.3	1.2	0.4	1.0		
	PrecStd	3.5	1.2	1.3	0.9	0.7	1.1	1.3	1.5	1.1	1.0	0.8	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	42.4	50.9	66.0	78.7	82.9	88.7	85.7	77.0	60.6	46.9	39.4		
		MCWB	36.7	35.7	39.9	48.8	55.5	60.0	63.0	62.2	58.5	50.3	42.3	35.3		
	2%	DB	38.4	38.7	46.2	58.4	73.0	78.2	82.6	81.0	71.5	55.0	41.3	36.4		
		MCWB	35.5	34.5	37.5	44.4	52.4	57.9	61.1	61.4	56.9	47.2	38.2	34.0		
	5%	DB	36.1	36.5	43.0	53.5	68.8	74.3	78.2	76.7	66.2	51.2	38.5	34.3		
		MCWB	34.1	33.2	35.7	41.5	51.1	56.1	59.7	59.5	54.5	45.1	35.9	32.8		
	10%	DB	34.1	34.3	40.2	49.8	64.4	70.1	74.0	72.2	61.4	47.9	36.3	32.6		
		MCWB	32.6	31.8	34.5	39.4	49.8	54.5	58.2	57.9	51.8	43.1	34.3	31.3		
	0.4%	WB	38.1	37.4	41.0	49.6	57.7	62.2	64.8	65.0	60.3	52.5	42.2	36.4		
		MCDB	40.4	39.9	47.7	64.2	72.7	77.1	82.4	80.1	73.1	58.2	45.7	38.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.9	35.3	38.7	45.4	54.6	59.8	62.8	62.5	57.5	48.9	38.9	34.5		
		MCDB	37.5	37.8	44.2	56.4	68.5	73.5	78.3	77.1	68.3	53.4	40.8	35.7		
	5%	WB	34.4	33.7	36.8	42.5	52.5	57.6	61.0	60.7	55.3	46.0	36.4	33.1		
		MCDB	36.0	36.0	41.7	51.2	65.6	70.4	74.9	73.9	64.3	50.0	38.0	34.3		
	10%	WB	32.7	32.0	35.0	40.5	50.4	55.8	59.2	58.9	53.1	43.5	34.4	31.4		
		MCDB	34.3	34.2	39.5	48.0	62.3	67.3	71.4	70.3	59.8	47.2	35.9	32.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	16.1	18.1	19.8	23.2	22.0	22.3	22.7	19.8	13.4	9.7	10.6		
		MCDBR	10.3	14.6	20.3	27.7	32.8	31.1	32.4	31.7	29.3	19.9	10.4	9.3		
		MCWBR	8.7	11.1	13.1	15.5	15.1	13.2	13.2	13.8	15.5	13.0	8.2	8.3		
		MCDBR	9.8	12.8	18.4	24.7	27.9	26.3	27.8	28.5	26.0	17.5	9.8	9.1		
	5% WB	MCWBR	8.5	10.1	12.7	14.6	13.5	11.9	12.2	12.9	14.3	12.3	8.2	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.237	0.259	0.274	0.311	0.336	0.346	0.340	0.337	0.303	0.289	0.261	0.222		
	taud		2.303	2.306	2.296	2.277	2.445	2.452	2.505	2.518	2.612	2.593	2.335	2.270		
	Ebn at Noon		231	266	288	290	288	285	285	279	278	256	216	209		
	Edh at Noon		16	23	30	36	33	33	31	29	23	18	16	13		
(44) All-Sky Solar Radiation	RadAvg		186	469	871	1424	1700	1802	1732	1470	956	494	203	137		
	RadStd		19	53	98	82	120	173	154	104	82	56	24	16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page